

Taewoong Yoon

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EDUCATION

Seoul National University (SNU)

Ph.D. in Physics · Advisor: Prof. Hyunyoung Choi · Seoul, Korea

Mar 2019 – Feb 2027 (expected)

Pohang University of Science and Technology (POSTECH)

B.S. in Physics · Pohang, Korea

Mar 2014 – Feb 2019

RESEARCH EXPERIENCE

Research Assistant, Korea Institute of Science and Technology (KIST)

Feb 2026 – Present

Advisor: Dr. Junghyun Lee

- Estimated defect densities from DEER data as part of a collaborative project with the Awschalom group (University of Chicago).
- Developing Python-based control software for a spin measurement setup, including FPGA-based timing and AWG pulse generation.

Graduate Research Assistant, Seoul National University (SNU)

Aug 2019 – Present

Advisor: Prof. Hyunyoung Choi

- Built the group's first NV center measurement system, including the confocal microscope, microwave delivery, pulse sequencing, and Python-based instrument control and data acquisition.
- Enhanced the polarization of the dark P1 electron spin bath 740-fold over thermal equilibrium by iterative Hartmann–Hahn transfer from NV centers, and characterized its collective coherence and disorder.
- Extended the NV–P1 protocols to P1–P1, probing interactions between hyperfine-resolved P1 subgroups and demonstrating polarization transfer between them.
- Developed a method to identify NV crystallographic axes from spatially varying microwave fields, enabling vector magnetometry without a calibrated bias field.
- Built a femtosecond laser-writing system with SLM-based aberration correction for site-controlled creation of color centers in diamond and hBN.

PUBLICATIONS

* Equal contribution. † Corresponding author.

1. **T. Yoon**, D. Kim, J. Lee[†], and H. Choi[†], “Polarization transfer between hyperfine-resolved P1 subgroups in diamond.” *In preparation*.
2. M. M. McMillan^{*}, H. Kim^{*}, J. C. Marcks, **T. Yoon**, B. Pingault, N. Delean, F. J. Heremans, H. Seo[†], J. Lee[†], and D. D. Awschalom[†], “Quantum decoherence in a mixed-dimensional spin bath.” *In preparation*.
3. **T. Yoon**, S. Oh, J. Lee[†], and H. Choi[†], “Mesoscopic spin coherence in a disordered dark electron spin ensemble,” arXiv:2602.17074 (2026). *Under review, Physical Review Applied*.
4. **T. Yoon**, M. Cha, D. Kim, and H. Choi[†], “Identifying NV center axes via spatially varying microwave fields for vector magnetometry,” *Applied Physics Letters* **126**, 144002 (2025).

PRESENTATIONS

1. **APS Global Physics Summit**, Denver, USA, Mar 2026. (*oral*)
“Polarization of P1 electron spin bath via repetitive Hartmann-Hahn transfer”
2. **KPS Spring Meeting**, Daejeon, Korea, Apr 2024. (*oral*)
“Probing and controlling many-body dipolar interactions between electron spins in diamond”
3. **Optics and Photonics Congress (OPC)**, Jeju, Korea, Aug 2023. (*oral*)
“Precise calibration of magnetic field vector in a diamond nitrogen-vacancy center ensemble”
4. **Conference on Lasers and Electro-Optics (CLEO)**, San Jose, USA, May 2022. (*poster*)
“Deterministic creation of single nitrogen-vacancy center in diamond using femtosecond laser writing”
5. **Optics and Photonics Congress (OPC)**, Jeju, Korea, Jul 2021. (*oral*)
“Aberration correction of femtosecond lasers for deterministic quantum emitters in nitrogen-vacancy diamonds”

TECHNICAL SKILLS

Coherent spin control — Pulse-sequence design on AWGs, TTL pulse generators, and FPGAs, including IQ-modulated microwave pulse shaping and synchronization of laser and photon-counting gates.

Optical measurement and control — Scanning confocal microscopy, gated and time-correlated photon counting, photoluminescence spectroscopy, AOM double-pass laser pulse control, femtosecond laser writing with SLM-based aberration correction.

Instrument control and data acquisition — Measurement automation in Python and LabVIEW.

Simulation and modeling — Spin dynamics and Monte Carlo simulations in Python (QuTiP), electromagnetic field modeling (COMSOL).

Fabrication — Photolithography, RIE, thermal evaporation, 2D material exfoliation and transfer.

Design — PCBs, photomasks, and machined mechanical parts (Altium, KLayout, AutoCAD, Fusion 360).

TEACHING EXPERIENCE

Teaching Assistant, Seoul National University

Electrodynamics I (Spring 2021) · Computational Physics (Fall 2020) · Electromagnetic Waves and Optics (Spring 2020) · Physics Lab. 2 (Summer 2019, Fall 2019) · Physics Lab. 1 (Spring 2019)

HONORS AND AWARDS

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| – BK Excellent Teaching Assistant Award | Jul 2020 |
| – BK Frontier Fellowship | Mar 2019 – Aug 2019 |
| – National Scholarship for Science and Engineering | Mar 2014 – Feb 2018 |